

Reinforcement Learning for Optimizing RAG for Domain Chatbots

발표자 : 조수빈, 자연어처리팀

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Introduction

- RAG (Retrieval Augmented Generation)
 - Retrieval Model, Answer Generation Model(LLM) 으로 구성
 - Retrieval Model은 input에 관련된 context를 가져옴
 - input과 검색된 context가 적절한 prompt와 함께 LLM에 입력되어 답변 생성
- RAG 기반 chatbot의 Challenges
 - Retrieval Model이 관련 context를 가져오지 못하는 경우, 생성된 답변이 부정확하거나 정보가 없을 수 있음
 - 유료 LLM의 경우, 입력 및 출력 토큰 수에 따라 비용이 청구됨
 - context 유지를 위해 대화 history를 LLM에 전달해야함
 - 입력 토큰이 많아지면 LLM의 정확도가 떨어지거나 hallucinations이 발생할 수 있음

RAG for FAQ chatbots

- Training in-house embedding model

- Top-k의 가장 관련성 높은 FAQ를 검색하는 과정
 1. query와 FAQ를 인코딩
 2. FAQ 내의 question과 answer을 연결하여 각 FAQ의 벡터를 생성
 3. cosine similarity를 사용하여 벡터들 사이의 유사도를 기반으로 FAQ rank
- Models
 - 공개적으로 이용 가능한 임베딩 모델(e5-base-v2)
 - in-domain 데이터와 infoNCE loss를 사용하여 e5-base-v2 모델을 fine-tuning
- Datasets
 - credit card application process에 관한 72개 FAQ로 구성된 데이터셋
 - ChatGPT를 사용해서 생성된 Question paraphrases 추가 (English, Few Hinglish)
 - FAQ answer를 기반으로 답변할 수 있는 query들을 Manual tagging한 작은 데이터 세트를 추가
 - train : 3.5k / valid : 1k / test : 1014개의 query 사용

RAG for FAQ chatbots

- InfoNCE loss

$$l_{i,j} = \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2B} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

- $\text{sim}(z_i, z_j)$: z_i 와 z_j 사이의 cosine similarity
- B : Batch size
- τ : temperature

RAG for FAQ chatbots

- Training in-house embedding model

Model	English		Hinglish	
	top-1	top-3	top-1	top-3
e5-base-v2	0.82	0.91	0.71	0.87
triplet-loss	0.90	0.93	0.84	0.89
infoNCE	0.97	0.98	0.94	0.95

Table 1: Retrieval accuracy comparison of public vs in-house embedding models

Model	In-domain	OOD
e5-base-v2	0.84	0.82
finetuned with InfoNCE	0.85	0.56

Table 2: Comparison of top-1 cosine similarity scores for in-domain vs OOD queries

RAG for FAQ chatbots

- Prompt for ChatGPT

- Prompting

1. contextual QnA에 적합한 프롬프트 추가
2. JSON 형식으로 결과를 얻기 위해 in-context 예시 추가
3. post 프롬프트 추가
 - 예: 'Instruction: Answer only using the information available.'know" 답변 제공
1. OOD query에 대해서는 "do not know"라고 답변하도록 설정

- 실험 설정

- 모든 실험에서 $k = 3$ 으로 설정
- ChatGPT 호출 시 마지막 두 대화를 history로 유지
 - history에는 이전 query, 생성된 chatbot 답변, FAQ 컨텍스트 포함

RAG for FAQ chatbots

- Evaluation

- GPT-4는 chatbot의 답변 품질을 ‘좋음/나쁨’으로 평가하고 평가 이유를 설명
- manual verification을 통해 GPT-4의 평가를 재확인, 100% 정확도를 가짐

query	is there annual fee for card
bot response	Yes, there is an annual fee of INR 500 for the credit card. However, this fee will be waived off if your annual spends exceed INR 3.5 Lakhs.
gpt4 eval	Good
gpt4 reason	The answer is directly addressing the query and providing the relevant information from the FAQ context. The answer matches with the information in the FAQ context, stating that there is an annual fee of INR 500 for the the credit card, which will be waived off if the annual spends exceed INR 3.5 Lakhs.

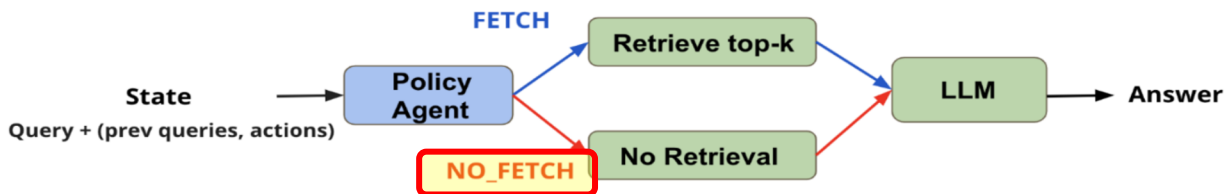
Table 3: GPT-4 evaluation for an in-domain query

query	hows weather today
bot response	I'm sorry, but I don't have information about the weather. How can I assist you with the credit card?
gpt4 eval	Good
gpt4 reason	The answer admits that it does not have information about the weather and asks how it can assist with the the credit card. The answer is relevant to the query and meets the answerability criteria.

Table 4: GPT-4 evaluation for an OOD query

Optimizing RAG for FAQ chatbots

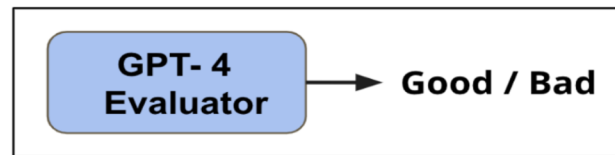
- Reinforcement learning for optimizing RAG



- [NO_FETCH] 의 시나리오

1. Follow up queries
2. Queries referring to the same FAQ
3. OOD queries

Reward model



Input: Query, Answer, top-k FAQ context, Conversation history

query	how much cashback on swiggy
policy action	[NO_FETCH]
bot response	You will get 1.5% unlimited cashback on Swiggy with the credit card.
gpt4 eval	Bad
gpt4 reason	The given answer is incorrect. According to the FAQ context, the credit card offers 4% unlimited cashback on preferred partners, including Swiggy. The answer provided 1.5% cashback, which is not accurate.

Table 5: GPT-4 evaluation for an in-correct policy action

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Action	GPT-4 eval	Reward
FETCH	-	0.1
NO_FETCH	Good	2
NO_FETCH	Bad	-1

Table 6: Reward shaping: converting GPT-4 evaluations to numeric rewards.

s_0	[CLS] is there annual fee [SEP]
a_0	[FETCH]
r_0	0.1
s_1	[CLS] is there annual fee [SEP] [FETCH] can you reduce it [SEP]
a_1	[NO_FETCH]
r_1	2.

Table 7: Sequence of (State, Action, Reward) for a sample chat session with two queries

$$G_t = \sum_{k=0}^N \gamma^k r_{t+k+1}$$

- γ : discount factor
- N : chat session에서 질문 수

Optimizing RAG for FAQ chatbots

- Reinforcement learning for optimizing RAG

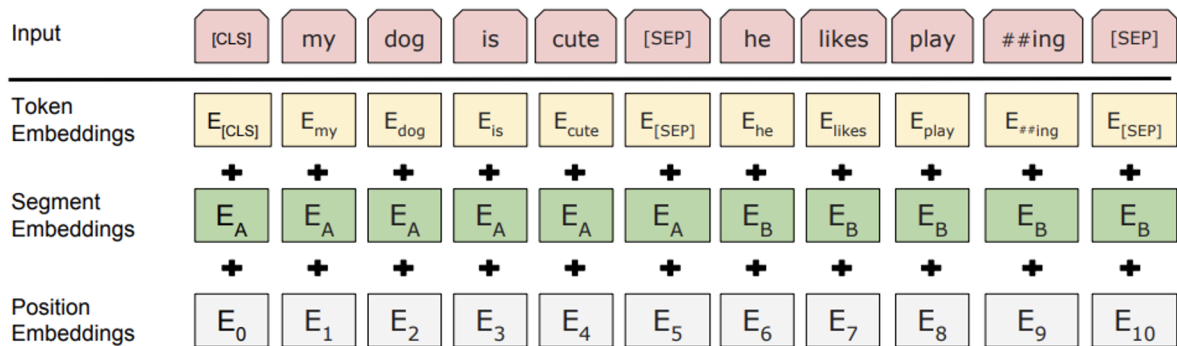


Figure 2: BERT input representation. The input embeddings are the sum of the token embeddings, the segmentation embeddings and the position embeddings.

$$l_t = -\log \pi_{\theta}(a_t|s_t)G_t - \lambda H(\pi_{\theta}(a_t|s_t))$$

- $\pi_{\theta}(a_t|s_t)$: 정책 네트워크가 주어진 state s_t 에서 action a_t 를 선택할 확률
- λ : 엔트로피 손실 가중치
- $H(\pi_{\theta}(a_t|s_t))$: 주어진 정책에 대한 엔트로피

Optimizing RAG for FAQ chatbots

- Similarity threshold on top-1 score (SimThr)

- Top-1 cosine similarity score가 threshold보다 높은 경우
 - 직접 FAQ 답변을 출력
- Top-1 cosine similarity score가 threshold보다 낮은 경우
 - 일반적인 RAG 파이프라인 실행

Optimizing RAG for FAQ chatbots

- Results

Setting	# tokens	saving	Acc.
All fetch	167885	-	0.989
SimThr	147147	~12%	0.989
SimThr + policy	115259	~31%	1.

Table 8: Comparison of token savings with different experimental settings

query	what cashback on i will get on swiggy
policy action	[FETCH]
bot answer	You will get 4% unlimited cashback on Swiggy with the credit card.
human eval	Good
query	what about pvr
policy action	[NO_FETCH]
bot answer	You will get 4% unlimited cashback on PVR with the credit card.
human eval	Good
query	how much on curefit
policy action	[NO_FETCH]
bot answer	You will get 4% unlimited cashback on Curefit with the credit card.
human eval	Good

Table 9: Policy model predictions on a sequence of queries referring to the same FAQ

Optimizing RAG for FAQ chatbots

- Comparison with gpt-2 as policy model

Setting	# tokens	saving	Acc.
SimThr + policy (gpt-2)	124924	~25%	1.
SimThr + policy (BERT)	115259	~31%	1.

Table 10: Comparison of token saving with a open gpt-2 as policy model

Conclusion

- in-house embedding Model Train
 - in-domain data, infoNCE loss 사용
 - ranking accuracy와 OOD query detection에서 좋은 성능
- LLM에 전달되는 토큰 수를 최적화
 - 강화학습 기반 최적화
 - Policy Model을 훈련시켜 context를 검색할지 여부 결정
 - GPT-4 이용하여 Reward 제공
 - Similarity threshold on top-1 score (SimThr)

→ 정확도 개선, token saving